

Practical 8

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In [1]: from sklearn.datasets import load_iris
        from sklearn.model_selection import train_test_split, cross_val_predict
        from sklearn.metrics import classification_report, roc_auc_score, confusion_matrix
        from sklearn.neural_network import MLPClassifier
        from sklearn.preprocessing import LabelBinarizer
        from sklearn.metrics import classification_report, roc_auc_score

        # Load the Iris dataset
        iris = load_iris()
        X_clf, y_clf = iris.data, iris.target

        # Split the data
        X_train, X_test, y_train, y_test = train_test_split(X_clf, y_clf, test_size=0.3, random_state=42)

        # Define and Train ANN model
        model = MLPClassifier(hidden_layer_sizes=(64, 32), max_iter=500, random_state=42)
        model.fit(X_train, y_train)

        # Evaluate on test data
        y_pred = model.predict(X_test)
        print(classification_report(y_test, y_pred))

        # Compute confusion matrix
        cm = confusion_matrix(y_test, y_pred)
        print("Confusion Matrix:")
        print(cm)
```

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# Compute AUC
lb = LabelBinarizer()
y_test_bin = lb.fit_transform(y_test)
y_pred_prob = model.predict_proba(X_test)
auc = roc_auc_score(y_test_bin, y_pred_prob, multi_class='ovr')
print(f"AUC Score: {auc}")
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	19
1	1.00	1.00	1.00	13
2	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

Confusion Matrix:

```
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

AUC Score: 1.0